

## EDUCATION

---

### Georgia State University – Atlanta, GA Bachelor of Science in Computer Science | Expected April 2026

- Relevant Coursework: Data Structures, Algorithms, Systems Programming, Operating Systems, Computer Organization, Software Engineering, Software Development, Mobile App Development, Web Programming, Cybersecurity Fundamentals, Data Science Fundamentals

Certificates of Git Essential Training: The Basics | September 2025

## WORK EXPERIENCE

---

### Freelance

**Software Engineer** | Georgia | June 2024 – Present

- Developed **6+ full stack web applications** for local businesses using **HTML, CSS, JavaScript, PHP, and SQL** delivering scalable, user-friendly, and production ready software.
- Deployed websites to production, using **Vercel** hosting, **MySQL** for database configuration, **Git** based version control, and domain setup.
- Improved product usability, accessibility, and performance, directly contributing to an average of **20%** increase in user engagement through client projects.
- Gathered requirements through conversations with clients and translated business needs to technical solutions demonstrating strong communication and product thinking skills.
- Implemented authentication systems, including flows for logging in and creating accounts, password hashing, session management, and input validation to prevent security risks.

### Select Management Resources

**Software Engineer Intern** | Alpharetta, Georgia | January 2023 – July 2023

- Developed **Python** and **JavaScript** scripts to streamline troubleshooting workflows, reduce repetitive operational tasks, and improve incident response efficiency.
- Designed and maintained structured technical documentation, system guides, and process workflows, improving onboarding efficiency by **20%**.
- Investigated application, system, and network issues through log analysis, ticket data review, and root cause analysis, contributing to a **25%** reduction in recurring incidents.
- Collaborated with infrastructure and operations teams to identify system reliability gaps and implement tooling and process improvements.
- Debugged and resolved complex software and system issues using structured problem solving, scripting-based validation, and logical decomposition.
- Produced internal reports and dashboards to track issue trends and performance metrics, enabling data driven operational decisions.

## PROJECTS

---

### Atlas: Social Fitness & Nutrition Platform (Flutter, Dart, Firebase, REST APIs, AI APIs)

- Developed a large mobile application that combines fitness tracking, nutrition logging, Artificial Intelligence driven planning, and social media functionality.
- Implemented secure authentication, user profiles, posts, likes, comments, and follower relationships using Firebase Authentication, Firestore, Storage, and security rules.
- Integrated multiple third-party REST APIs to support exercise discovery, workout logging, macro tracking, barcode food lookup, and nutrition data retrieval.
- Built an Artificial Intelligence powered assistant using OpenAI API to generate personalized workout and meal plans through structured prompt engineering based on user constraints (equipment, dietary preferences, environment).
- Created modular application pages including Feed, Tracking (Workouts & Nutrition), Artificial Intelligence Assistant, and Profile pages with scalable data models and role-based access control.

### Sliding Puzzle Web Game (HTML, CSS, JavaScript, PHP, MySQL)

- Developed a full stack sliding puzzle game with secure user authentication, session management, and persistent gameplay statistics.
- Implemented account registration and login using PHP sessions, password hashing, and prepared SQL statements to ensure secure credential handling.
- Built dynamic JavaScript game logic supporting multiple difficulty levels, timers, move tracking, power ups, animations, and real time state updates.
- Designed and integrated a MySQL database and analytics dashboard to track user performance metrics including completion time, moves, power up usage, achievements, and historical results.
- Tested the application with 12 users and deployed it on a school hosted server, refining UI/UX and gameplay balance based on feedback.

### Traffic Sign Image Recognition System (Python, PyTorch, scikit-learn, OpenCV, Pandas, NumPy, Torchvision, Matplotlib)

- Built an image recognition and classification models for **43** traffic sign classes using the **GTSRB** dataset, training, evaluating, and optimizing multiple machine learning models.
- Developed a **Neural Network** using **ResNet18** and **Transfer Learning** achieving up to a **94%** test accuracy through stage-based training, and fine tuning.
- Implemented multiple baseline models including HOG + SVC, k-NN, and Random Forest models, and through hyperparameter tuning was able to improve model test accuracies from as low as **77%** to as high as **97%** improving overall model accuracies by **10%-20%**.
- Evaluated models based statistical analysis using precision, recall, F1-score, and matplotlib for confusion matrices visualization to analyze class level errors to better tailor optimizations and improve accuracies
- Engineered data preprocessing and augmentation pipelines, including normalization and standardization of the image features.

## TECHNICAL SKILLS

---

- Programming Languages: Python, Java, JavaScript, C++, C, C#, Dart, HTML, CSS, SQL, PHP, YAML
- Technologies: GitHub, Git, Emulators / Simulators, Xcode, Microsoft Office, Google Suite, MySQL, DBeaver, Firebase, Linux, Unix, Command Line, Unit Testing, Data Structures, Algorithms, Flutter
- Core Skills: Object Oriented Programming, Database Management, Full stack Development, App Development, Web Development, Debugging, Troubleshooting, Automation Scripting, UI/UX, Server-Side Programming, Version Control, REST APIs, AI API Integrations, Prompt Engineering, State Management, Authentication, Authorization, Mobile UI Architecture